

Executive Summary

Irreversible Arithmetic Communication Capability

Overview

We present a working communication mechanism that enables practical transmission of structured data in a form that is unreadable without proper initialization parameters and readable with them, despite the absence of a classical cryptographic secret, reversible cipher, or stable ciphertext.

The system operates by deliberately exploiting irreversible numerical behavior in real-world computational environments. It is not compliant with classical cryptographic or information-theoretic models — and this is a design feature, not a defect.

What This Is (in operational terms)

- A method to **transmit documents or structured data** such that:
 - unauthorized recipients cannot recover usable content,
 - authorized recipients can reconstruct meaningful data with high fidelity.
 - The mechanism has been **implemented, tested, and demonstrated** on real hardware.
 - It functions under realistic conditions including rounding errors, precision loss, and numerical instability.
-

What Makes It Different

- **No stable ciphertext**
Data is not represented as a fixed symbolic object that can be stored, replayed, or algebraically analyzed.
 - **No classical secret to brute-force**
Initialization parameters do not provide a binary “correct/incorrect” test, eliminating standard key search strategies.
 - **No correctness oracle**
Attackers cannot algorithmically determine whether they are close to success.
 - **Irreversibility by design**
The computational process destroys information pathways typically exploited in cryptanalysis.
-

Practical Consequences

- Standard cryptanalytic approaches (bruteforce, differential, algebraic, statistical) **do not apply directly**, because their prerequisites are absent.

- Attack cost cannot be meaningfully estimated using existing models.
- Automation of attacks is severely constrained due to the lack of objective success criteria.

This does **not** imply absolute security.

It implies **uncertainty asymmetry** in favor of the defender.

Limitations (Known and Managed)

- No formal security proof in the classical sense.
- Some degradation of reconstructed data may occur at local (block-level) granularity.
- System behavior is sensitive to numerical environment and must be tested per platform.

These limitations are explicit, observable, and operationally bounded.

Why This Matters

This capability addresses scenarios where:

- Formal cryptographic assurances are less critical than **practical non-readability**.
- Adversaries are resourceful but constrained by time, tooling, and uncertainty.
- Systems must function in degraded, noisy, or non-ideal computational environments.

It provides **a new option**, not a replacement, in the secure communication toolbox.

Status

- Functional prototype implemented.
 - Empirical validation performed.
 - Further development would focus on:
 - operational hardening,
 - integration constraints,
 - red-team style adversarial testing.
-

Bottom Line

This is a working communication capability that remains effective precisely because it operates outside the assumptions that attackers rely on.

If the organization requires **formal proofs**, this is not the right tool.

If the organization requires **that unauthorized parties cannot read transmitted content in practice**, this capability warrants serious evaluation.